

Part II – Database Design & Implementation

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1. Logical Data Model – Entity Relationship Diagram

The clinic database contains several entities that capture users, medical staff, patients, scheduling and appointment information. A high level summary of each entity and its primary characteristics is shown below.

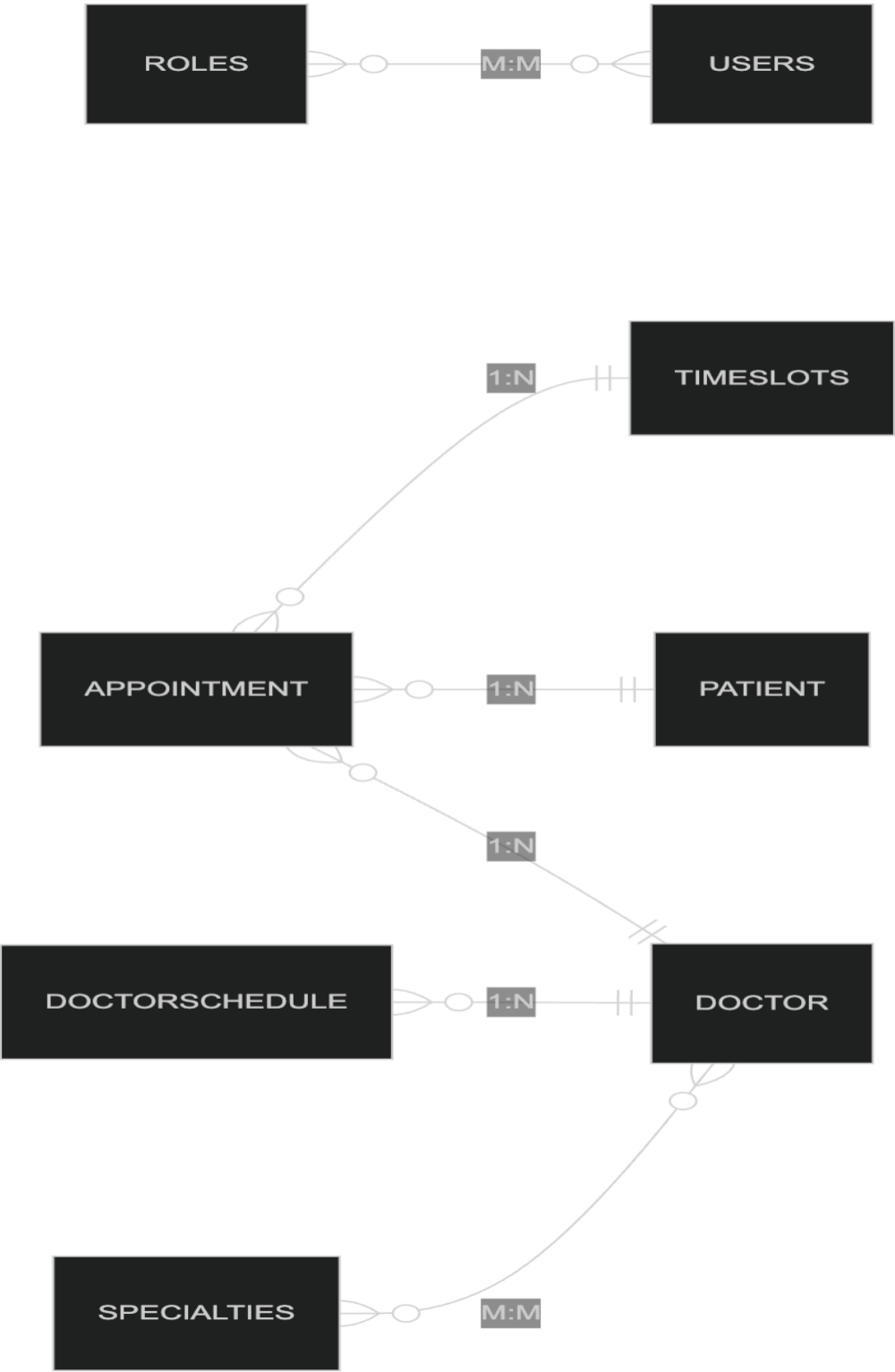
Entity	Primary Key(s)	Important attributes/notes
Users	Id	Username, password hash, created/modified timestamps.
Roles	Id	Name of the role, created/modified timestamps.
UserRoles	UserId, RoleId	Junction table linking users to roles (M:N).
Patient	Id	First & last names, insurance number (unique), date of birth, phone number.
Doctor	Id	First & last names, license number (unique).
Specialties	Id	Name of a medical specialty.
DoctorSpecialties	DoctorId, SpecialtyId	Junction table linking doctors to specialties (M:N).
DoctorSchedule	Id	DoctorId, day of week, work

		start/end times.
TimeSlots	Id	Hour of day, minute of hour; unique half-hour combinations.
Appointment	Id	Date, notes, FKs to patient/doctor/timeslot; unique per doctor/date/slot.

Relationships and cardinalities

- Users ↔ Roles – many-to-many using UserRoles.
- Doctors ↔ Specialties – many-to-many using DoctorSpecialties.
- Doctor → DoctorSchedule – one doctor may have multiple schedule entries.
- Doctor → Appointment – one-to-many; unique (DoctorId, Date, TimeSlotId) prevents double booking.
- Patient → Appointment – one-to-many.
- Appointment → TimeSlots – many appointments may reuse a slot (different doctors/dates).

The following figure illustrates the logical entity relationship model for the clinic database:



2. Physical Data Model – Relational Data Model

Relational tables and key attributes:

Table name	Key columns & data types	Other columns and constraints
Users	Id INT PRIMARY KEY	Username VARCHAR(32) UNIQUE; PasswordHash; timestamps.
Roles	Id INT PRIMARY KEY	RoleName VARCHAR(64); timestamps.
UserRoles	UserId INT, RoleId INT composite PK	FKs to Users and Roles with ON DELETE CASCADE.
Patient	Id INT PRIMARY KEY	InsuranceNumber UNIQUE with CHECK; DOB; phone; timestamps.
Doctor	Id INT PRIMARY KEY	LicenseNumber UNIQUE; timestamps.
Specialties	Id INT PRIMARY KEY	Name UNIQUE.
DoctorSpecialties	DoctorId, SpecialtyId composite PK	FKs with ON DELETE CASCADE.
DoctorSchedule	Id INT PRIMARY KEY	DoctorId FK; DayOfWeek CHECK; WorkStart/End; timestamps.
TimeSlots	Id INT PRIMARY KEY	Hour/Minute CHECK; UNIQUE(hour, minute).
Appointment	Id INT PRIMARY KEY	Date; Notes; FK Patient/Doctor/TimeSlot; UNIQUE (DoctorId, Date, TimeSlotId).

Domain level constraints

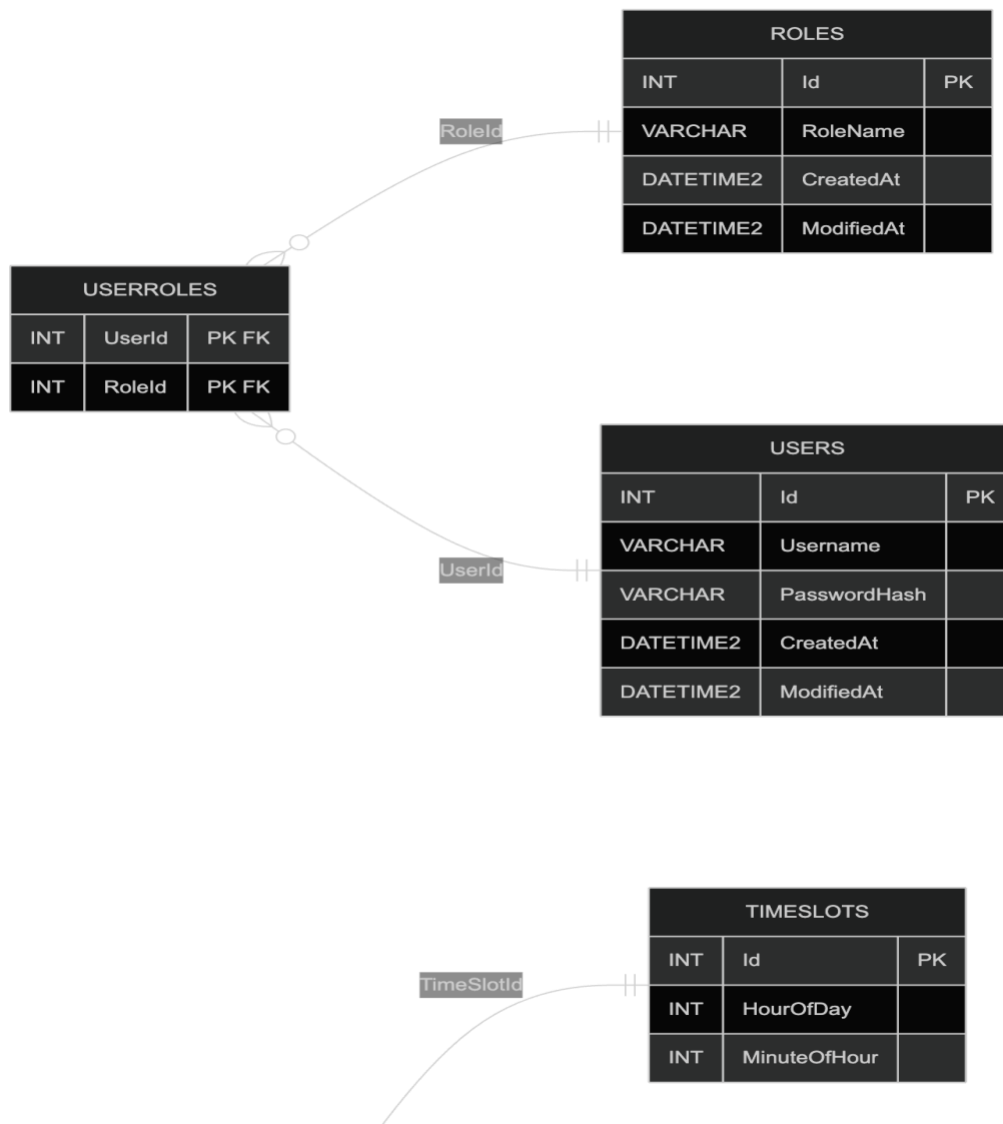
- Insurance number formats validated with CHECK.
- DayOfWeek limited to Sunday–Saturday.
- TimeSlots restrict HourOfDay (0–23) and MinuteOfHour (0–59).

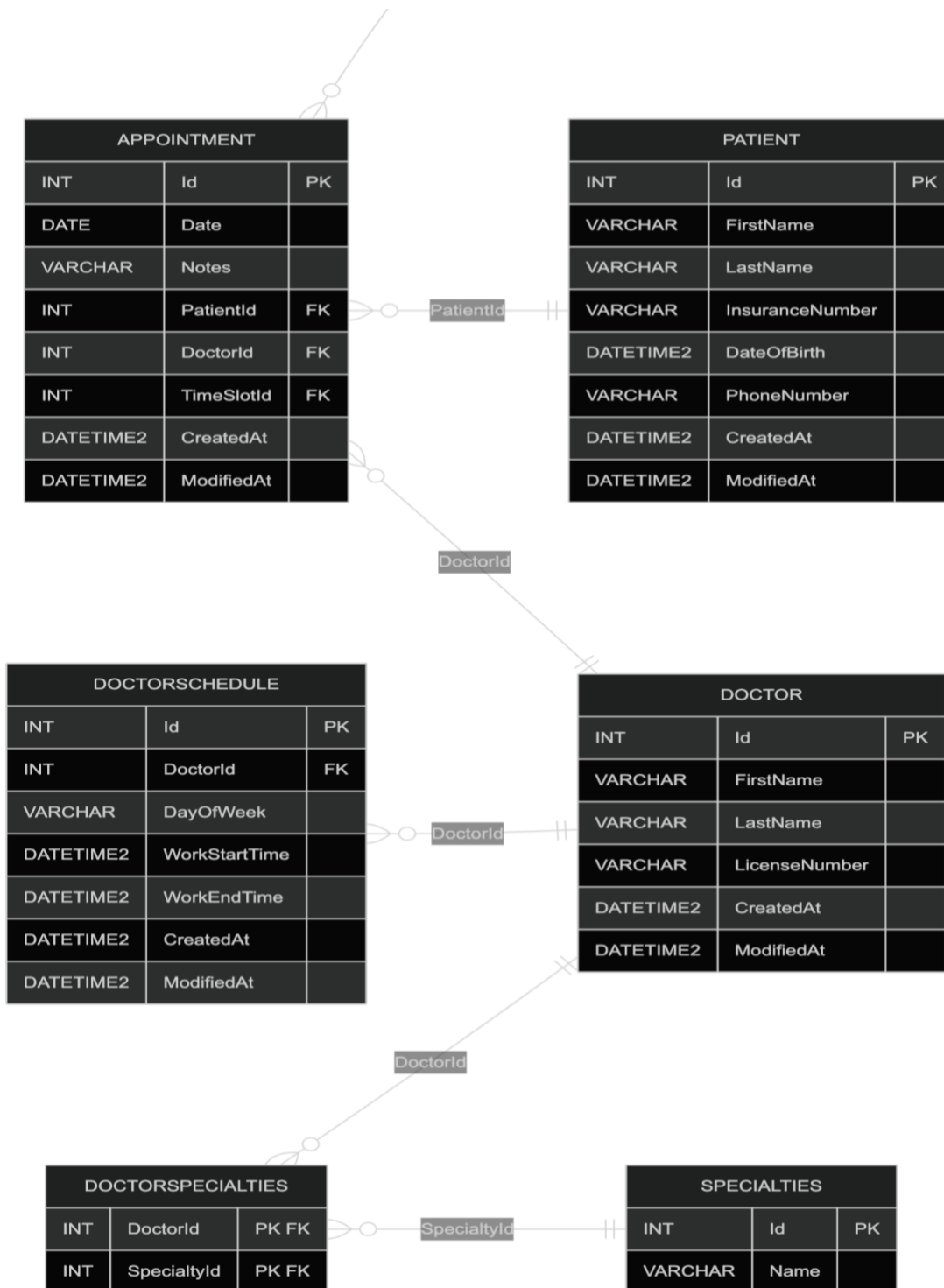
- Appointment unique constraint prevents double-booking.

Table and relationship constraints

- Primary keys uniquely identify records.
- Foreign keys enforce referential integrity with cascade where needed.
- Unique constraints enforced on usernames, role names, doctor license numbers, patient insurance numbers, and slot combinations.

RDM Diagram





3. SQL Implementation

The database has been implemented on Microsoft SQL Server using Transact-SQL. The script defines tables, constraints, sample data, views, and stored procedures such as scheduling, booking, cancelling and retrieving appointment details. All objects are created through a single SQL file (HealthCareClinicDB.sql).